Expression of *IL24*, *ABCA9* and *ABCB7* in Relation to Maternal Age, BMI and Preeclampsia

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November 11, 2026

1 Methods

1.1 Data Source

Data were obtained from a nested case–control gene expression study within the Vitamin D Antenatal Asthma Reduction Trial (VDAART) [1], available from the Gene Expression Omnibus (accession GSE85307). Peripheral blood gene expression was measured at 10–18 weeks of gestation using the Affymetrix Human Gene 1.0 ST Array. The initial dataset contained 157 samples.

1.2 Data Processing and Sample Size

Expression samples were matched to participant information using the GEO sample identifier (geoID), and all 157 samples were successfully matched. Probe IDs were mapped to gene symbols, and expression values from multiple probes for the same gene were averaged. Covariate values were converted from text to numeric or categorical form. Maternal BMI was categorized using World Health Organization cut-points (underweight < 18.5, normal weight 18.5–24.9, overweight 25–29.9, obesity ≥ 30 kg/m^2). No participants had missing data for pregnancy outcome, maternal age or BMI, so no samples were removed and the final sample included 157 participants (110 normal pregnancies and 47 preeclampsia cases).

1.3 Statistical Analysis

Three genes were selected for analysis: *IL24*, *ABCA9* and *ABCB7*. For each gene, a histogram of expression, a scatterplot of expression against maternal age with a LOESS trend line, and a boxplot of expression by pregnancy outcome and BMI category were generated. Associations were assessed with Spearman correlation (maternal age), Wilcoxon rank-sum tests (pregnancy outcome) and Kruskal–Wallis tests (BMI category), with $p < 0.05$ considered statistically significant. Participant characteristics were summarized as mean (SD) for continuous variables and n (%) for categorical variables, stratified by pregnancy outcome. A heatmap of ten genes was generated using row-scaled (z-score) expression with hierarchical clustering of genes and participants (Euclidean distance, complete linkage). Finally, a dumbbell plot was used to compare mean z-scored expression of the same ten genes between pregnancy outcome groups.

1.4 Software

All analyses were performed in R version 4.6.1 [2]. Data were processed using dplyr (v1.2.1) [3] and tidyr (v1.3.2) [4]. Figures were created with ggplot2 (v4.0.3) [5] and pheatmap (v1.0.13) [6], and the report was generated with knitr (v1.51) [7].

1.5 Code Availability

All code is available at `https://github.com/YOUR-USERNAME/YOUR-REPO`.

2 Results

2.1 Participant Characteristics

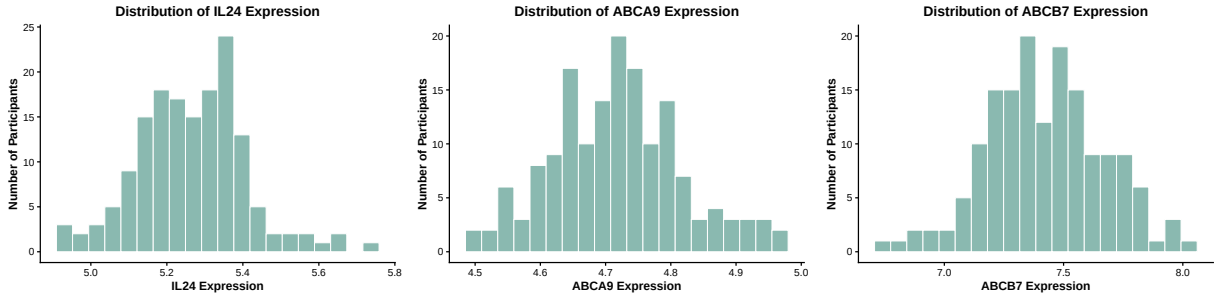
Maternal age was similar between groups, but women who developed preeclampsia had lower vitamin D levels, delivered earlier, and were more often classified as obese than women with normal pregnancies (Table 1).

Table 1: Participant characteristics stratified by pregnancy outcome. Continuous variables are presented as mean (SD) and categorical variables as n (%). Vitamin D was measured as serum 25-hydroxyvitamin D at baseline.

Characteristic	Overall (n = 157)	Normal (n = 110)	Preeclampsia (n = 47)
Maternal age (years), mean (SD)	26.4 (5.0)	26.6 (5.1)	25.8 (4.9)
Vitamin D (ng/mL), mean (SD)	23.0 (13.1)	24.4 (14.5)	19.7 (8.3)
Gestational age at delivery (weeks), mean (SD)	38.8 (2.2)	39.3 (1.0)	37.8 (3.5)
Maternal BMI category, n (%)			
Underweight	5 (3.2)	4 (3.6)	1 (2.1)
Normal weight	48 (30.6)	36 (32.7)	12 (25.5)
Overweight	47 (29.9)	39 (35.5)	8 (17.0)
Obesity	57 (36.3)	31 (28.2)	26 (55.3)

2.2 Distribution of Gene Expression

Expression of *IL24*, *ABCA9* and *ABCB7* was unimodal and approximately symmetric, with a slight right tail for *IL24* and *ABCA9* (Figure 1).

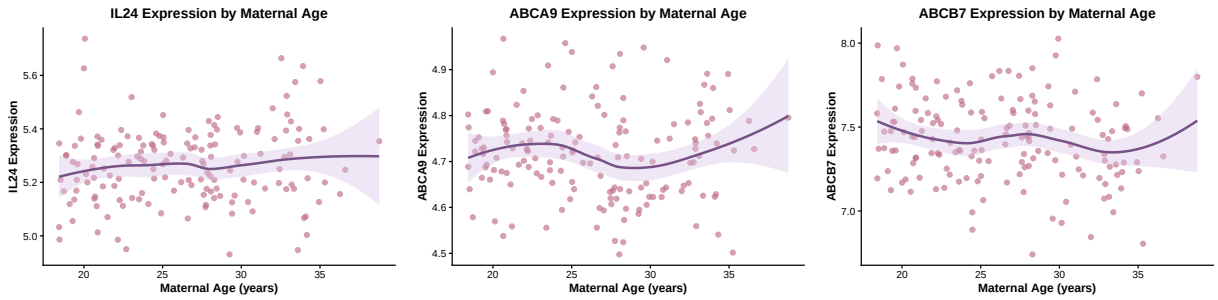


(a) *IL24* **(b)** *ABCA9* **(c)** *ABCB7*

Figure 1: Distribution of expression of (a) *IL24*, (b) *ABCA9* and (c) *ABCB7* across all 157 participants.

2.3 Gene Expression and Maternal Age

None of the three genes was significantly associated with maternal age. *IL24* ($\rho = 0.12$, $p = 0.14$) and *ABCA9* ($\rho = -0.04$, $p = 0.60$) showed no meaningful relationship, while *ABCB7* showed a weak, non-significant negative trend ($\rho = -0.15$, $p = 0.057$) (Figure 2).

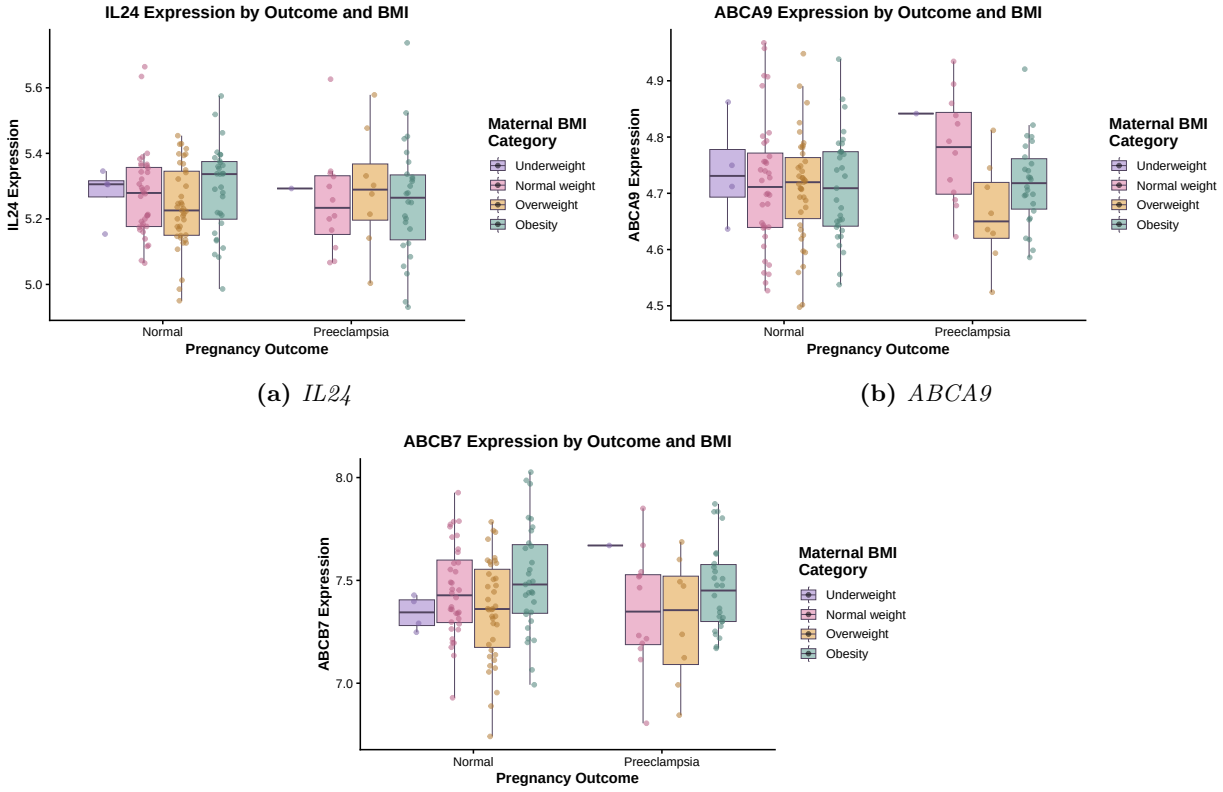


(a) *IL24* **(b)** *ABCA9* **(c)** *ABCB7*

Figure 2: Gene expression versus maternal age for (a) *IL24*, (b) *ABCA9* and (c) *ABCB7*. Each point represents one participant; the line shows a LOESS fit with a 95% confidence band.

2.4 Gene Expression by Pregnancy Outcome and BMI

Expression of *IL24*, *ABCA9* and *ABCB7* did not differ between normal and preeclamptic pregnancies ($p = 0.60$, 0.43 and 0.81, respectively) or across maternal BMI categories ($p = 0.78$, 0.49 and 0.15, respectively) (Figure 3).



(a) *IL24* **(b)** *ABCA9*

(c) *ABCB7*

Figure 3: Gene expression by pregnancy outcome and maternal BMI category for (a) *IL24*, (b) *ABCA9* and (c) *ABCB7*. Boxes show the median and interquartile range; points represent individual participants.

2.5 Clustering of Ten Genes

Hierarchical clustering of ten genes did not separate participants by pregnancy outcome or maternal BMI category, as both tracking bars were intermixed across clusters (Figure 4).

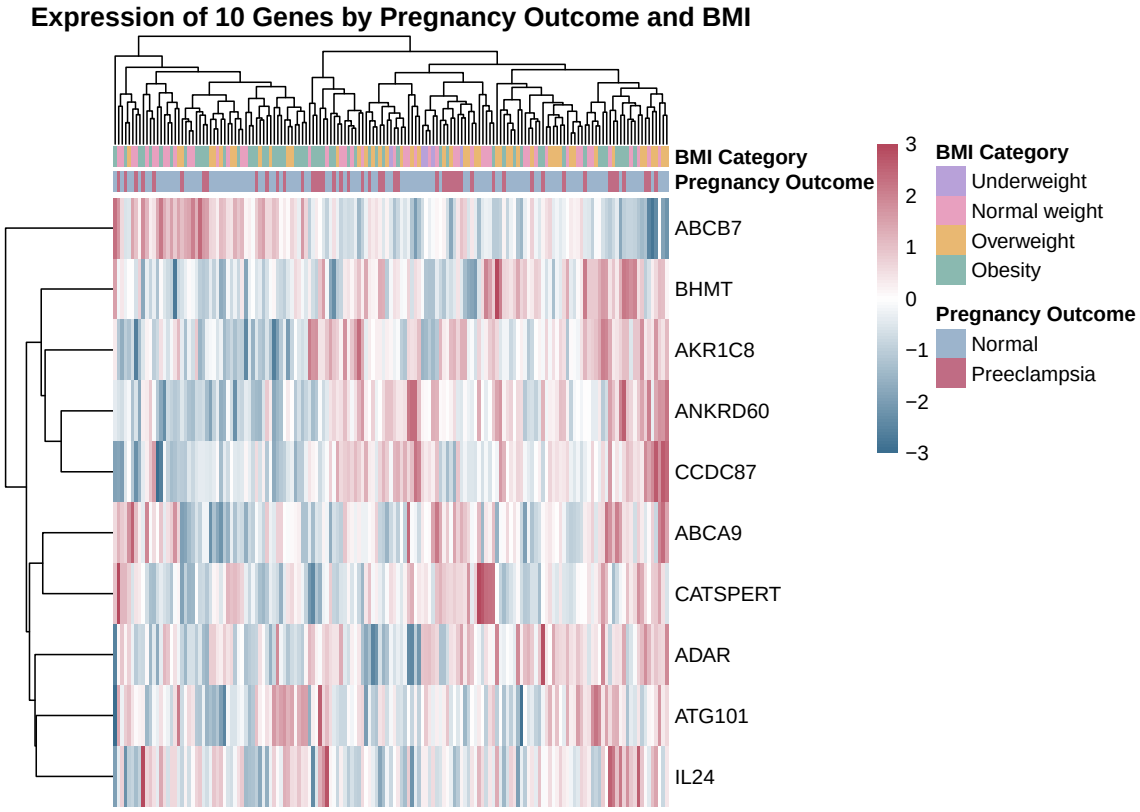


Figure 4: Heatmap of row-scaled (z-score) expression for ten genes across 157 participants. Genes (rows) and participants (columns) were clustered hierarchically using Euclidean distance and complete linkage. Tracking bars indicate pregnancy outcome and maternal BMI category. Red indicates higher and blue lower expression relative to each gene's mean; colors are capped at ± 3 SD.

2.6 Differences in Mean Expression Between Outcome Groups

Differences in mean expression between outcome groups were small for all ten genes (less than 0.4 standard deviations), and the 95% confidence intervals of the two groups overlapped for every gene. *ATG101*, *ANKRD60* and *AKR1C8* showed the largest increases in preeclampsia, whereas *CATSPERT* and *BHMT* were slightly lower (Figure 5).

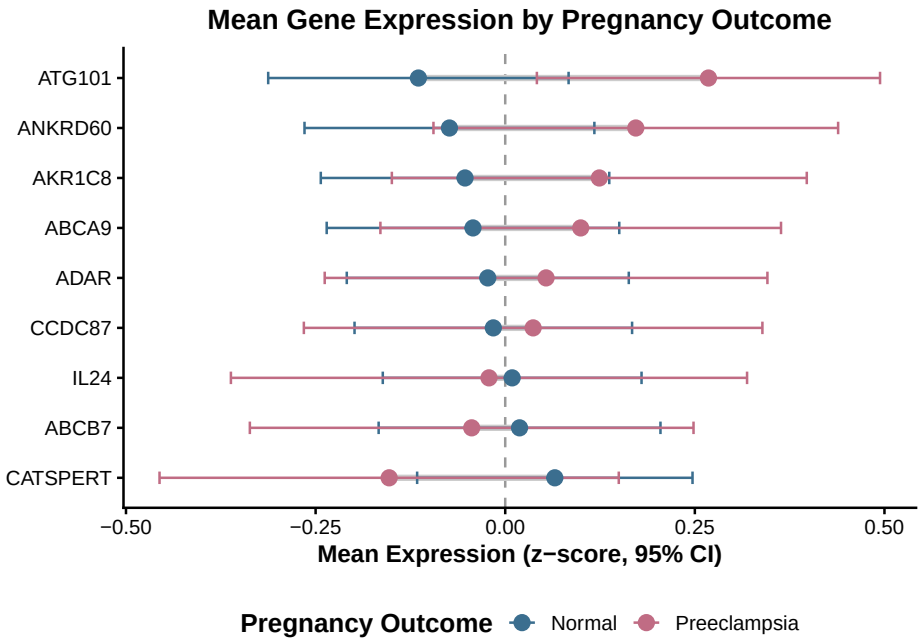


Figure 5: Dumbbell plot of mean z-scored expression for ten genes in normal ($n = 110$) and preeclamptic ($n = 47$) pregnancies. Points show group means with 95% confidence intervals; grey bars connect the two group means for each gene. Genes are ordered by the difference between groups.

References

[1] Hooman Mirzakhani, Augusto A. Litonjua, Thomas F. McElrath, et al. Early pregnancy vitamin D status and risk of preeclampsia. *Journal of Clinical Investigation*, 126(12):4702–4715, 2016.

[2] R Core Team. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria, 2026.

[3] Hadley Wickham, Romain François, Lionel Henry, Kirill Müller, and Davis Vaughan. *dplyr: A Grammar of Data Manipulation*, 2026. R package version 1.2.1.

[4] Hadley Wickham, Davis Vaughan, and Maximilian Girlich. *tidyr: Tidy Messy Data*, 2025. R package version 1.3.2.

[5] Hadley Wickham, Winston Chang, Lionel Henry, Thomas Lin Pedersen, Kohske Takahashi, Claus Wilke, Kara Woo, Hiroaki Yutani, Dewey Dunnington, and Teun van den Brand. *ggplot2: Create Elegant Data Visualisations Using the Grammar of Graphics*, 2026. R package version 4.0.3.

[6] Raivo Kolde. *pheatmap: Pretty Heatmaps*, 2025. R package version 1.0.13.

[7] Yihui Xie. *knitr: A General-Purpose Package for Dynamic Report Generation in R*, 2025. R package version 1.51.